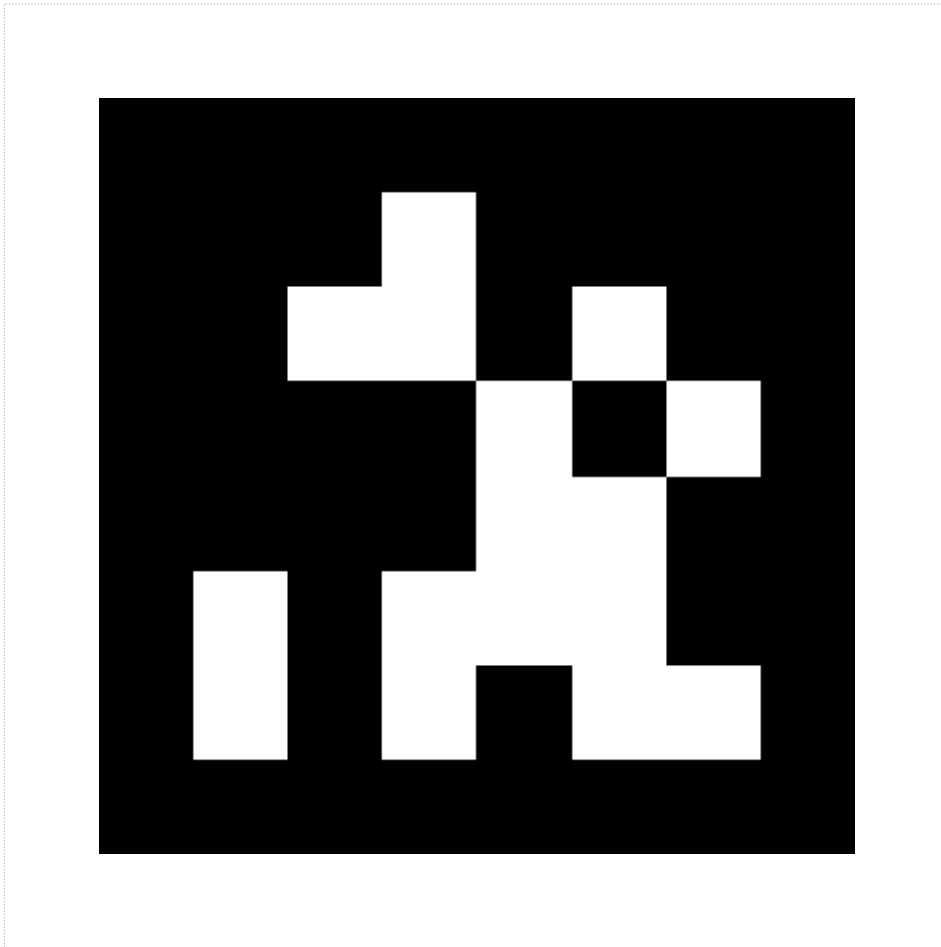


AprilTag 36h11 - hand-eye target (100 mm)

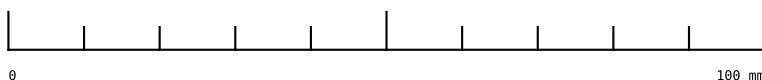
Print at 100% / actual size. Do NOT use 'fit to page' or 'shrink to fit'.



id = 0
100.0 mm

Hand-eye calibration target. Fix it to the floor or table ^{HAND-EYE / BASE} so it cannot move while the arm is moved through calibration poses.

- The black border is part of the tag: 100.0 mm is the OUTER black square.
- Keep the white margin around it clear (≥ 12.5 mm).
- Mount it flat. A bulge of 1 mm over 100 mm is already 0.6 deg of pose error.
- D405 at 300 mm sees this as ~225 px across: plenty for pose.



0

100 mm

<- must measure exactly 100 mm.
If not, the print was scaled.

AprilTag 36h11 - holder tags (10 mm)

Print at 100% / actual size. Cut along the dashed line to keep the quiet zone.



id = 1
slot 1



id = 2
slot 2



id = 3
slot 3



id = 4
slot 4



id = 5
slot 5



id = 6
slot 6



id = 7
slot 7



id = 8
slot 8



id = 9
slot 9



id = 10
slot 10

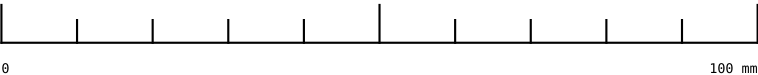


id = 11
MEAS.
HOLDER

Rack slot tags (id 1-10) and the measurement holder tag (id 11), 10.0 mm.
Every tag carries a distinct id, so a detection identifies which holder it is.

10 mm is usable from 70 to ~150 mm only (96 px down to 45 px on a D405).
Beyond ~200 mm detection becomes unreliable and pose is unusable.
That is fine for close-up localization; it is NOT enough for hand-eye,
which is why the 100 mm tag on page 1 exists.

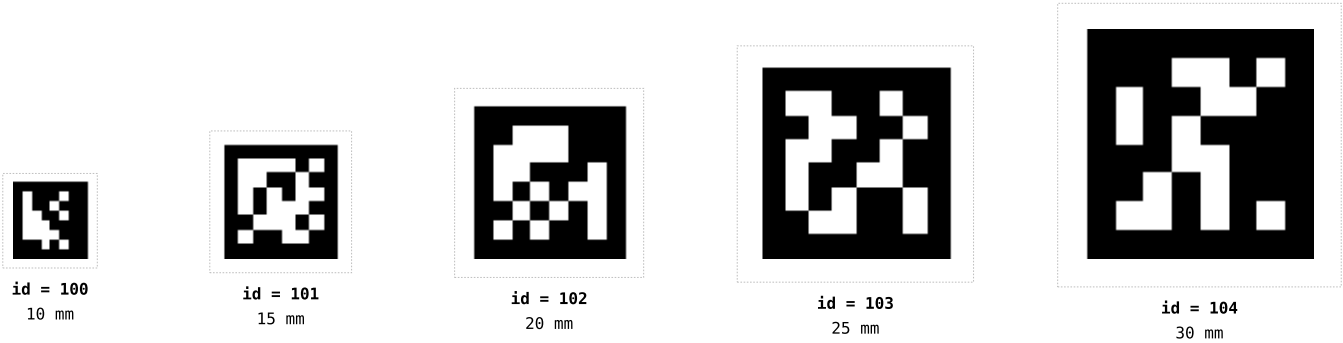
Each cell is 1.25 mm. Print on a 600 dpi laser printer or better;
inkjet bleed rounds the corners and biases the detected pose.
Matte paper. Glossy or laminated surfaces blow out under the D405 projector.



<- must measure exactly 100 mm.
If not, the print was scaled.

AprilTag 36h11 - size trial strip

Print at 100% / actual size. Use this to choose the holder tag size empirically.



Trial strip. Before committing to 10 mm, tape these where a holder tag would go and check at the distance the wrist camera actually observes from.

Expected pixel size on a D405 (fx ~674 px):

dist	10mm	15mm	20mm	25mm	30mm
70mm	96	144	193	241	289
100mm	67	101	135	169	202
150mm	45	67	90	112	135
200mm	34	51	67	84	101
300mm	22	34	45	56	67

Rule of thumb: >= 30 px to detect, >= 60 px before trusting the pose.
Pick the smallest size that still clears 60 px at your working distance.

